



DATA PROJECT

Customer Shopping Behavior Analysis

An end-to-end analysis of **3,900 transactions** using Python, PostgreSQL, and Power BI — uncovering spending patterns, segments, and strategic insights.

Dataset at a Glance

3,900

Total Purchases

Transactional records analyzed

18

Data Columns

Demographics, purchases & behavior

37

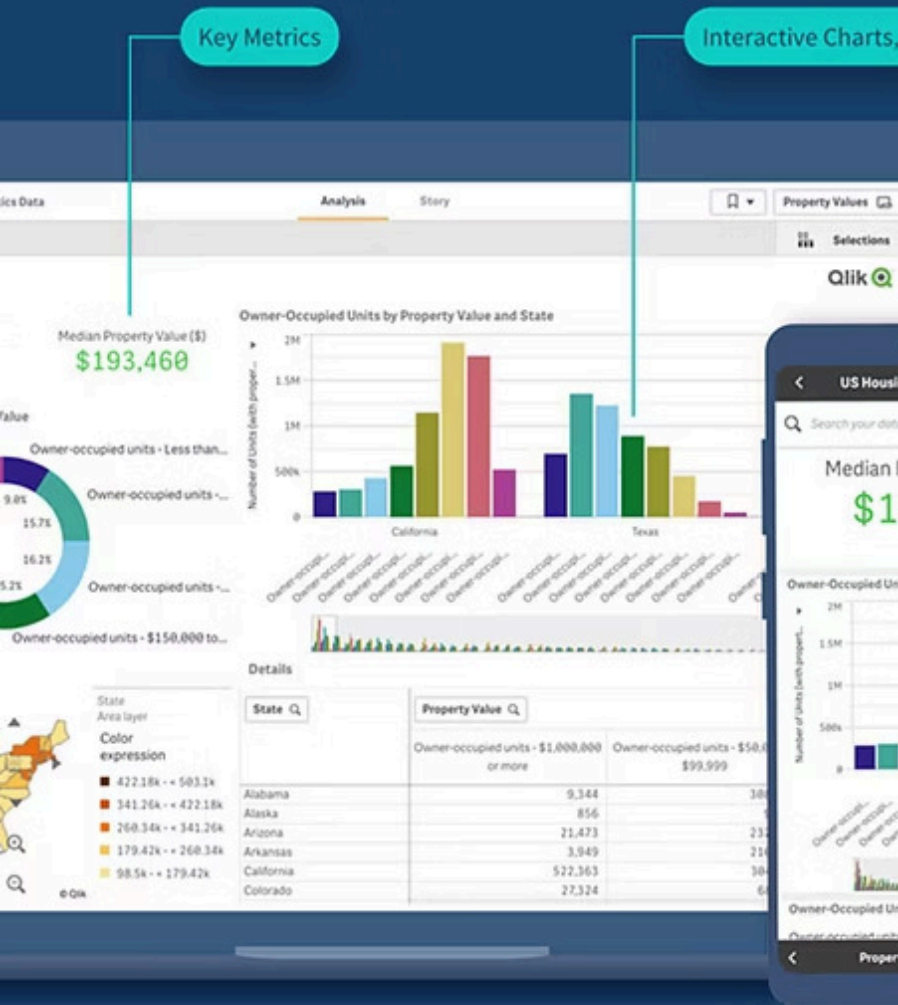
Missing Values

All in Review Rating column

\$59.76

Avg. Purchase

Mean purchase amount (USD)

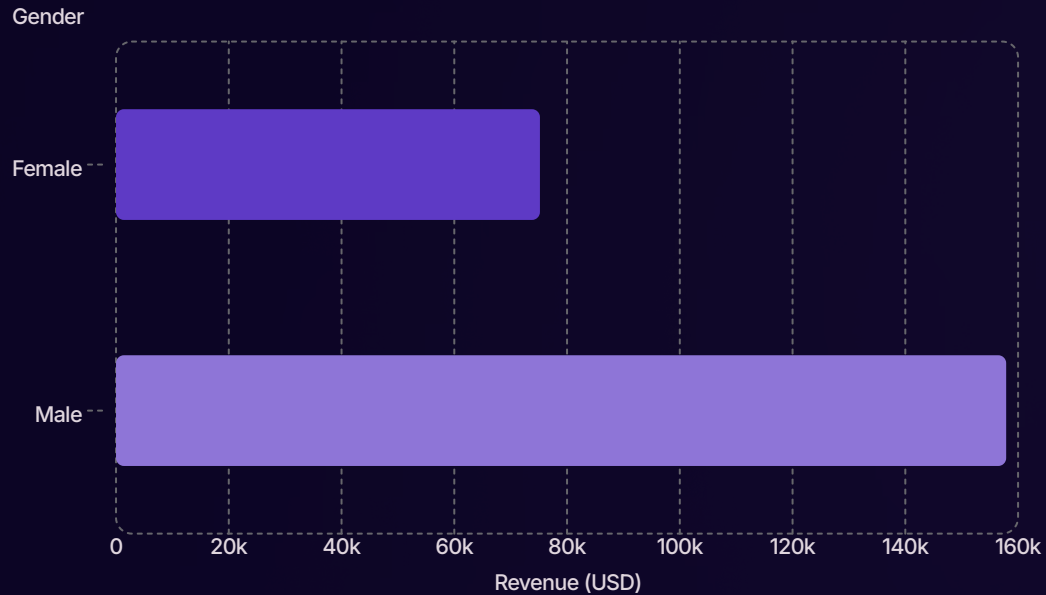


Data Preparation in Python



Cleaned and engineered in Python before loading into PostgreSQL for SQL-based analysis.

Revenue by Gender & Discount Users

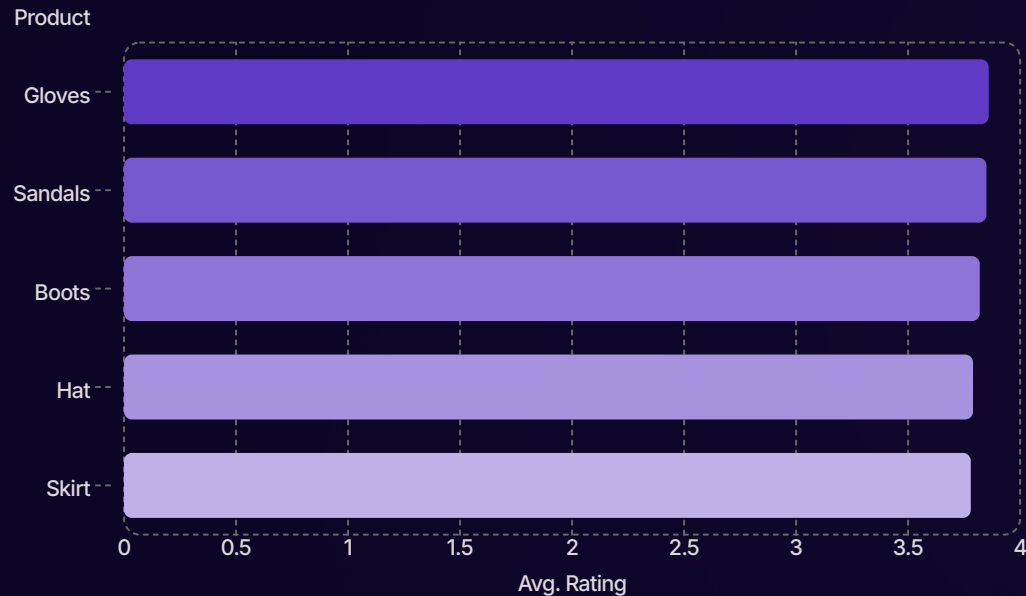


Key Findings

Male customers generated **2× the revenue** of females — \$157,890 vs. \$75,191.

839 discount users still spent above the \$59.76 average — discount use ≠ low value.

Product Ratings & Shipping Behavior

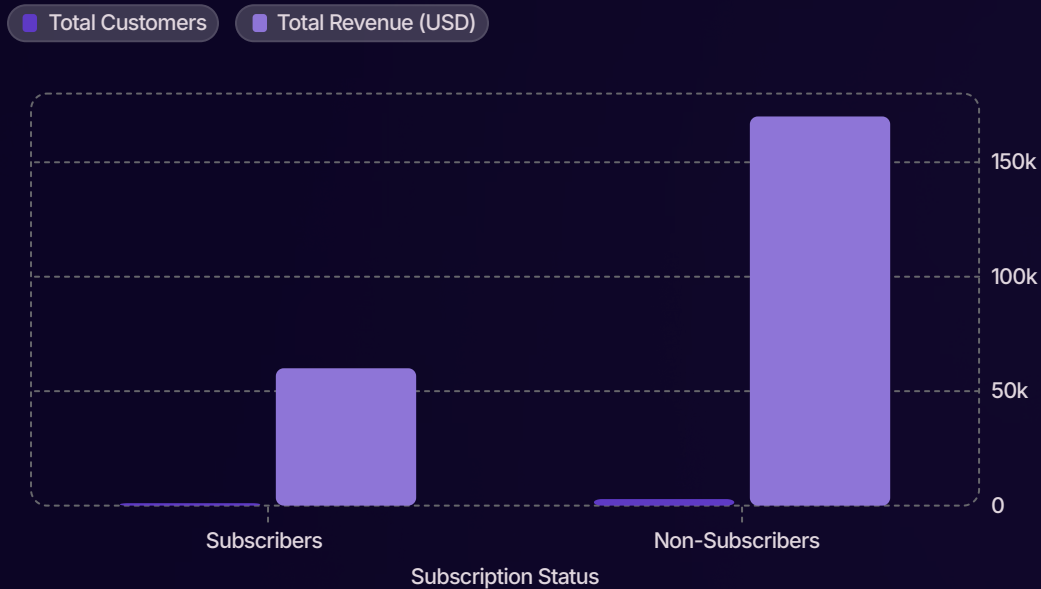


Ratings & Shipping

Overall avg. rating: **3.75** — top items cluster tightly (3.78–3.86).

Express shipping avg.: **\$60.48** vs. Standard **\$58.46** — a \$2.02 difference suggesting urgency correlates with higher basket values.

Subscribers vs. Non-Subscribers



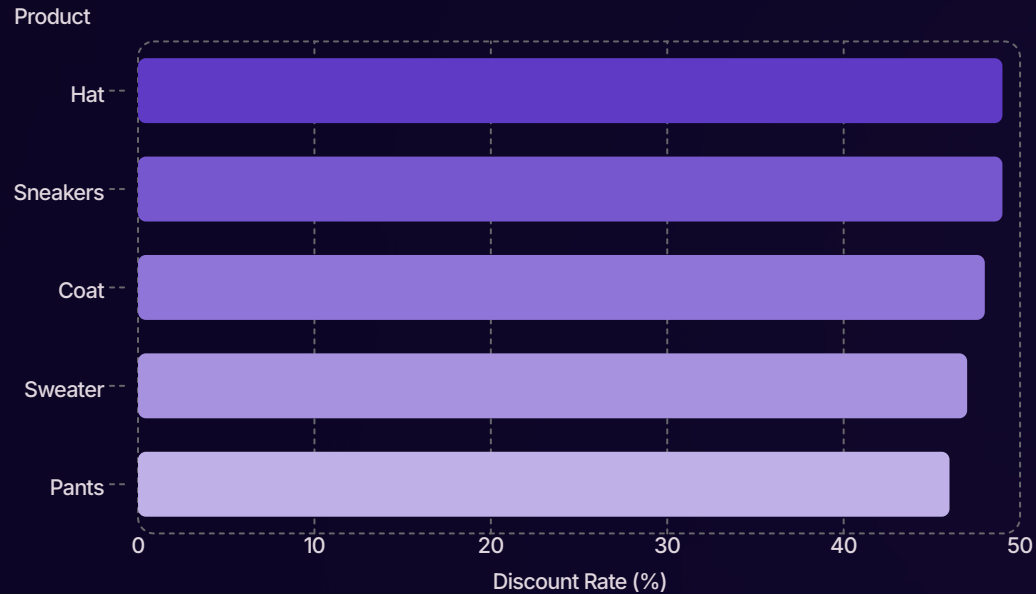
The Subscription Gap

Non-subscribers vastly outnumber subscribers (**2,847 vs. 1,053**) and drive most revenue.

Yet avg. spend is nearly identical — **\$59.49 vs. \$59.87**.

i Opportunity: convert the large non-subscriber base.

Discount-Dependent Products & Customer Segments



Customer Segments

Loyal

3,116 customers — high frequency

Returning

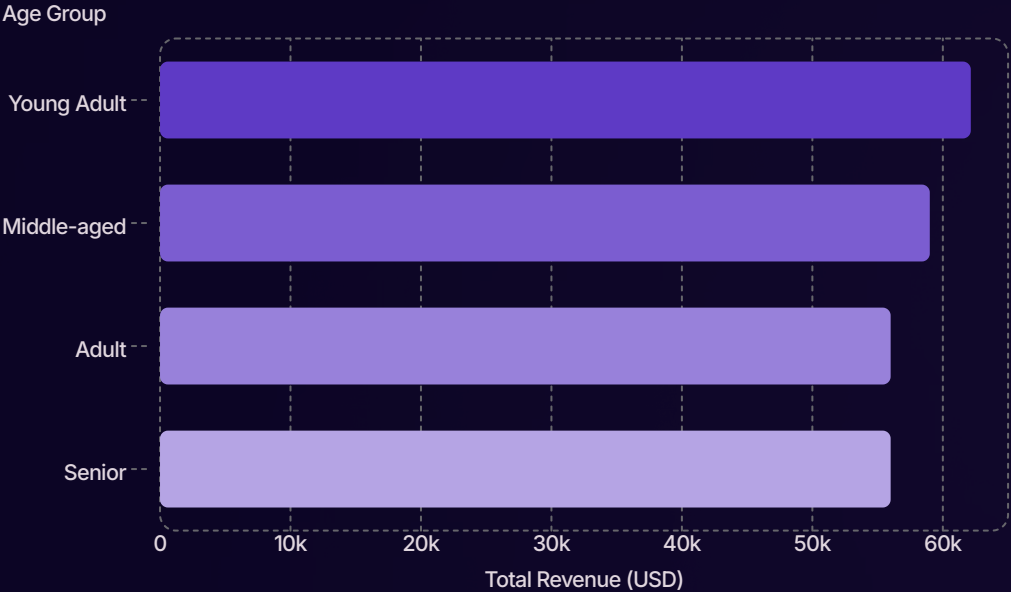
701 customers — growth potential

New

83 customers — retention target

Nearly **50%** of these product purchases involve a discount — margin risk to monitor.

Top Products by Category & Revenue by Age



Top Products per Category

Category	Product	Orders
Accessories	Jewelry	171
Clothing	Blouse	171
Footwear	Sandals	160
Outerwear	Jacket	163

Young Adults lead revenue at **\$62,143** — gradual decline across older groups.

Power BI Dashboard

Interactive dashboard enabling stakeholders to filter and explore data dynamically.

Key KPIs

3.9K customers · \$59.76 avg.
purchase · 3.75 avg. rating

Filters

Subscription, Gender, Category
(4 types), Shipping (6 options)

Visuals

Subscription donut, Revenue by Category, Revenue by Age Group



Business Recommendations



Boost Subscriptions

Convert 2,847 non-subscribers driving 73% of revenue.



Loyalty Programs

Reward repeat buyers to grow the Loyal segment (3,116 of 3,900).



Review Discount Policy

~50% of Hat, Sneakers, Coat purchases use discounts — balance sales with margin.



Product Positioning

Highlight top-rated (Gloves, Sandals, Boots) and best-sellers (Blouse, Jewelry, Jacket).



Targeted Marketing

Focus on Young Adults (\$62K revenue) and express-shipping users.